LITERATURE REVIEW

LITERATURE REVIEW 1

Title: A Framework for Online Academic Project Evaluation

Authors: Siddharth Mittal et al. (2025)

Source: International Journal for Research in Applied Science & Engineering Technology (IJRASET),Volume 13, Issue IV, April 2025

DOI: https://doi.org/10.22214/ijraset.2025.68337

Methodology :

The authors propose an online framework for managing and evaluating academic projects using a centralized digital platform. The system focuses on automating project evaluation processes, reducing manual intervention, and improving scalability. It discusses the use of emerging technologies such as artificial intelligence and plagiarism detection to maintain academic integrity. The framework also considers workflow optimization, resource allocation, scheduling, and privacy concerns.

Summary:

This paper addresses the limitations of traditional manual academic project evaluation methods and introduces a modern online evaluation framework. By digitizing project management activities, the proposed system improves transparency, efficiency, and consistency in assessment. The inclusion of plagiarism detection mechanisms helps ensure originality, while structured workflows support effective monitoring and evaluation of academic projects.

Advantages:

- Supports online and centralized academic project evaluation
- Reduces manual effort and administrative workload
- Improves transparency and standardization in assessments
- Incorporates plagiarism detection to ensure academic integrity
- Scalable framework suitable for large institutions
- Emphasizes workflow optimization and efficiency

Disadvantages:

- Role-based access for students, guides, and coordinators is not clearly defined
- Rubric-based evaluation is mentioned but not detailed
- AI usage is high-level without concrete implementation details
- Stage-wise approval and weekly progress tracking are not addressed
- Dataset and experimental validation are limited

LITERATURE REVIEW 2

Title: Designing a Web-Based Information System for Monitoring Final Projects
Authors: Ferry Hariyanto et al. (2025)
Source: International Journal of Engineering, Science and Information Technology (IJESTY), Volume 5, Issue 2, April 2025
DOI: https://doi.org/10.52088/ijesty.v5i2.799

Methodology:
The authors propose the design of a web-based information system aimed at monitoring and managing final-year student projects. The system replaces manual and fragmented project handling processes with a centralized digital platform. Unified Modeling Language (UML) is used to analyze system requirements and define workflows involving students, supervisors, examiners, and administrators. The approach focuses on improving project supervision, documentation management, and coordination through an integrated online system.

Summary:
This paper highlights the inefficiencies of conventional manual project monitoring methods, such as lack of transparency, poor communication, and difficulty in tracking student progress. The proposed web-based system enables centralized storage of project data and improves interaction between stakeholders. Although the study mainly emphasizes system design rather than full implementation, it demonstrates how digital platforms can enhance monitoring, coordination, and administrative efficiency in academic project management.

Advantages:
- Provides a centralized platform for monitoring student projects
- Improves coordination between students, supervisors, and administrators
- Reduces reliance on manual documentation and reporting
- Enhances transparency in project supervision
- Clearly defines system structure using UML modeling
- Supports systematic tracking of academic project activities

Disadvantages:
- Does not implement a complete working system for real-world validation
- Role-based access control is not clearly detailed
- Rubric-based or standardized evaluation mechanisms are not addressed

LITERATURE REVIEW 3

Title: Student Project Management System for Monitoring Undergraduate Projects
Author: Abiola O. Arowolo et al. (2024)

Source: Vol. 10, No. 3, Series, University of Ibadan, Nigeria

DOI: https://www.crossref.org/06members/50go-live.html

Methodology:
The proposed Student Project Management System (SPMS) is a web-based platform designed to manage and monitor undergraduate projects. The system supports proposal submission, project tracking, supervisor feedback, document management, and communication between students, supervisors, and coordinators. It follows a structured development lifecycle including requirements analysis, system design, implementation, testing, and documentation to ensure usability and reliability.

Dataset:
• Dataset Collection: Project-related academic data such as student profiles, project proposals, progress records, feedback, and submitted documents collected from the Computer Science Department.
• Dataset Processing: Data was stored and managed in a centralized MySQL database to support tracking, monitoring, and coordination.

Summary:
The paper "Student Project Management System for Monitoring Undergraduate Projects" presents a web-based solution aimed at improving the management of final-year undergraduate projects. The system enhances collaboration between students, supervisors, and coordinators by providing tools for project tracking, feedback management, and document organization. By centralizing project information and automating monitoring activities, the system improves administrative efficiency, project visibility, and overall student project outcomes.

Advantages:
• Enables continuous monitoring of undergraduate projects.
• Improves communication between students and supervisors.
• Centralized system enhances transparency and coordination.

Disadvantages:
• Limited to a single department case study.
• Does not emphasize standardized rubric-based evaluation.

LITERATURE REVIEW 4

Title: UMIT Project Management System

Author: Anita Morey et al. (2024)

Source: Blue Eyes Intelligence Engineering and Sciences Publication (BEIESP), International Journal of Innovative Technology and Exploring Engineering (IJITEE), Volume 13, Issue 8, Pages 29–35

DOI : https://doi.org/10.35940/ijitee.H9935.13080724

Methodology:

The UMIT Project Management System (PMS) is an automated web-based platform designed to manage academic projects conducted across multiple semesters and phases. The system digitizes core project activities such as synopsis submission, group formation, guide allocation, evaluation scheduling, and rubric-based assessment. Role-based dashboards are provided for administrators, faculty guides, and students to ensure structured evaluation and transparent mark calculation.

Dataset :

• Dataset Collection: Academic project data including student details, project phases, evaluation rubrics, and guide information collected from institutional records.

• Dataset Processing: Data was structured and digitized phase-wise to support automated evaluation and monitoring.

Summary:

The paper "UMIT Project Management System" proposes an automated solution to overcome the challenges of manual project evaluation at UMIT. Since academic projects span multiple semesters and phases with different rubric formats, manual handling becomes complex and error-prone. The proposed PMS streamlines project management and evaluation by automating workflows, improving accuracy, transparency, and efficiency. The system supports structured monitoring of Project I, II, and III, thereby enhancing academic quality and administrative control.

Advantages:

• Reduces manual workload in project evaluation and management.

• Improves accuracy and transparency through rubric-based automation.

• Centralized system enables efficient coordination among users.

Disadvantages:

• Requires institutional technical support for implementation.

• Initial digitization of records may be time-consuming.

LITERATURE REVIEW 5

Title: Fuzzy Intelligent System for Student Software Project Evaluation

Authors: Anna Ogorodova et al. (2025)

Source: International Journal of Modern Education and Computer Science (IJMECS), 2025

DOI: https://doi.org/10.5815/ijmecs.2025.04.02

Methodology:
The paper proposes a fuzzy logic–based system to evaluate student software projects. Evaluation criteria were identified through surveys involving student project teams and faculty members. Parameters such as code cleanliness, use of object-oriented concepts, and functionality were modeled using fuzzy sets. A fuzzy inference system processes these inputs to generate an overall project performance score.

Dataset:
Collection: Survey data from 31 student project teams and 3 faculty members
Processing: Conversion of qualitative evaluation criteria into fuzzy variables for automated assessment

Summary:
The study addresses the difficulty of fairly evaluating student software projects in large classes. By using fuzzy logic, the system converts subjective evaluation parameters into quantifiable scores. This approach reduces manual grading effort and ensures consistency in evaluation while providing timely feedback to students.

Advantages :

- Automates the evaluation of student software projects
- Reduces subjectivity and bias in manual grading
- Ensures consistent and transparent assessment
- Handles increased instructor workload effectively
- Provides quick feedback to students

Disadvantages:

- Evaluation is limited to a few technical parameters
- Requires expert knowledge to define fuzzy rules
- Does not cover full project lifecycle management
- No support for plagiarism detection
- Progress tracking and approval workflows are not included

LITERATURE REVIEW 6

Title : Project Monitoring with Decision Support System for the University Infrastructure Development Office

Authors : Abraham Simon (2023)

Source: Journal of Multidisciplinary Research and Development, Vol. 2 No. 1, 2023

DOI : https://doi.org/10.60008/thequest.v2i1.59

Methodology :
The study focuses on the development of an Online Project Monitoring System designed to improve project tracking and decision-making within a university's Infrastructure Development Office. The system was developed using the Agile Software Development Methodology, which emphasizes iterative development, continuous feedback, and incremental improvements. Each development cycle involved planning, design, implementation, testing, and evaluation to ensure that the system met user and organizational requirements. The system replaces traditional paper-based project monitoring methods with a centralized digital platform, enhancing data organization and accessibility. System evaluation was conducted using the ISO/IEC 25010 software quality model to assess functionality, usability, performance efficiency, security, reliability, maintainability, and portability.

Summary :
The paper presents the design and implementation of an online project monitoring system intended to address inefficiencies in manual project record-keeping within a university infrastructure context. By adopting an Agile development approach, the system was iteratively refined to ensure usability and functional adequacy. The digital platform enables better organization, tracking, and monitoring of ongoing projects, supporting informed decision-making by administrators. Evaluation results based on ISO/IEC 25010 standards indicate that the system satisfies essential software quality requirements and demonstrates potential to enhance operational efficiency and project oversight within academic institutions.

Advantages :
- Eliminates manual, paper-based project monitoring processes
- Improves organization and accessibility of project data
- Supports timely monitoring and administrative decision-making
- Ensures software quality through standardized ISO/IEC 25010 evaluation
- Enhances efficiency in managing university infrastructure projects

Disadvantage :
- Primarily focused on infrastructure development projects rather than academic or student-centric project management
- Limited discussion on long-term scalability and integration with other institutional systems
- Advanced analytics and automation features are not extensively addressed
- User training and change-management aspects are minimally discussed

LITERATURE REVIEW 7

Title: Web-based Projects Evaluation Management System

Authors: S. Al-Zoubi et al. (2016)

Source: Journal of Computer Science, Vol. 4, No. 11, 2016

DOI: https://doi.org/10.3844/jcssp.2008.916.921

Methodology :
The study adopted a design research methodology involving problem identification, system suggestion, development, and evaluation. It addressed inefficiencies in the manual evaluation of MSc final projects at the College of Arts and Sciences (CAS). A web-based solution was developed to allow supervisors and evaluators to submit project evaluation forms online, replacing traditional paper-based methods.

Summary :
The paper presents a web-based project evaluation management system developed to digitize the final project assessment process at CAS. By enabling online submission and storage of evaluation data, the system reduces administrative workload, saves time and resources, and improves accessibility and efficiency in academic project evaluation.

Advantages :
- Reduces paperwork and manual effort
- Saves time and administrative resources
- Improves efficiency in project evaluation

Disadvantages :
- Focuses only on project evaluation
- Does not support full project lifecycle management

LITERATURE REVIEW 8

Title : Prototype Development of Final Year Project Management System to Monitor Student's Performance

Authors : Raznida Isa et al. (2024)

Source : Journal of Advanced Research in Applied Sciences and Engineering Technology, 2024

DOI : https://doi.org/10.37934/araset.40.1.164173

Methodology : The study by Raznida Isa et al. focuses on the development of a prototype web-based Final Year Project Management System (FYPMS) to address limitations of manual project supervision. Using a dashboard-centric design, the system was developed following a structured software development life cycle, including requirement gathering, design and analysis, software development, testing, launch, and maintenance phases. The methodology involved user interviews and requirement analysis, which informed the system design that allows students, lecturers, and coordinators to interact efficiently. Users can log in using institutional credentials, and the system enables progress tracking and communication between stakeholders.

Summary : The paper by Raznida Isa et al. highlights the design and implementation of a centralized academic project management system that improves monitoring and supervision of final year student projects. The system replaces manual logbooks and reduces faculty workload by providing real-time progress visibility to supervisors and coordinators. It enables streamlined communication and project submission workflows, allowing lecturers to review and grade reports online. The use of a dashboard interface enhances the visibility of student performance trends and supports timely intervention where necessary.

Advantages :

- Replaces manual project tracking with an efficient online system
- Improves visibility of project progress and timeline adherence
- Supports multi-role functionality for students, lecturers, and coordinators
- Reduces administrative workload and improves supervision outcomes
- Enhances communication and report submission efficiency

Disadvantages :

- Prototype is limited to early implementation phases and may lack advanced features
- Does not explicitly include rubric-based evaluation or automated plagiarism checking.
- Focuses on supervision and monitoring rather than complete project lifecycle management
- Integration with academic performance systems and notifications is not fully addressed

LITERATURE REVIEW 9

Title: Hybrid Online-Offline Teaching Mode in Microcontroller Project-Based Learning

Authors: Danhua Cao et al. (2025)

Source: Proceedings of SPIE – Education and Training in Optics and Photonics (ETOP 2025), Volume 13706, 2025

DOI: https://doi.org/10.1117/12.3076806

Methodology:

The authors implemented a hybrid online–offline teaching model combined with project-based learning (PBL) for a microcontroller course. Classroom lectures were supported with online digital resources, and student performance was evaluated using a structured assessment system aligned with Outcome-Based Education (OBE). The approach focused on balancing online and offline activities to improve student engagement and learning outcomes.

Summary:

This paper explores how a hybrid teaching approach can enhance project-based learning in technical courses. By integrating online resources with offline classroom instruction, the system improves student motivation and learning effectiveness. The study emphasizes structured assessment and outcome-based education to ensure students acquire practical and theoretical skills relevant to industry requirements.

Advantages :

- Supports hybrid (online + offline) learning for project-based courses
- Improves student engagement and learning motivation
- Uses structured assessment aligned with Outcome-Based Education (OBE)
- Encourages hands-on project development
- Effective for post-pandemic academic environments

Disadvantages:

- Focuses on teaching methodology rather than project management systems
- No centralized software platform for project tracking
- Lacks role-based access for students, guides, and coordinators
- Does not include plagiarism detection
- No automated project evaluation or rubric-based grading system

LITERATURE REVIEW 10

Title: The Use and Management of Project Control Systems
Authors: Morris et al. (2025)
Source: Journal Article, 2025
DOI: 10.1016/0263-7863(87)90002-1

Methodology:
The study investigates project monitoring and control practices used in large-scale and complex projects through qualitative case analysis. The methodology includes examination of project plans, monitoring reports, and managerial decision records to understand how project performance is tracked over time. The research focuses on techniques for monitoring schedule, cost, and progress deviations and how feedback from monitoring activities supports corrective actions during project execution.

Summary:
This paper emphasizes the critical role of project monitoring and control in ensuring project success. Continuous monitoring enables early detection of deviations from planned objectives and supports timely managerial intervention. The study establishes that systematic tracking of project performance improves predictability, accountability, and coordination throughout the project lifecycle.

Advantages:
- Establishes foundational concepts for project monitoring and control
- Emphasizes continuous performance tracking and feedback
- Supports informed and timely decision-making
- Applicable to various types of projects across industries
- Provides theoretical grounding for modern monitoring practices

Disadvantages:
- Focuses on traditional monitoring approaches rather than automated systems
- Limited use of real-time data and digital tools
- Does not address collaborative or multi-user monitoring environments
- Lacks consideration of modern analytical and intelligent monitoring methods

LITERATURE REVIEW 11

Title: An Investigation of the Impact of Software Project Management on Software Development Success

Authors: Wallace et al. (2017)

Source: Journal Article, 2017

DOI: 10.1287/isre.1040.0039

Methodology:
The study employs a quantitative research methodology based on survey data collected from software development professionals involved in large-scale software projects. The research investigates the relationship between software project management practices and project success by analyzing factors such as planning effectiveness, risk management, scheduling, and coordination. Statistical modeling techniques are used to evaluate how management practices influence project outcomes, including quality, timeliness, and stakeholder satisfaction.

Summary:
This paper emphasizes the critical role of effective software project management in achieving successful software development outcomes. The findings indicate that structured planning, continuous monitoring, and proactive risk management significantly improve project performance. The study demonstrates that systematic management practices reduce project failures and enhance the predictability and quality of software development processes.

Advantages:
- Highlights the importance of structured management in software projects
- Demonstrates measurable impact of management practices on project success
- Supports proactive risk management and monitoring approaches
- Encourages disciplined planning and control mechanisms
- Applicable to both academic and industrial software projects

Disadvantages :
- Focuses primarily on industrial software development environments
- Does not address system automation or digital project management tools
- Lacks emphasis on role-based system access and collaboration platforms
- Does not include academic project supervision or evaluation mechanisms

LITERATURE REVIEW 12

Title: Online Integrated Platform for Project Management

Author: et al. (2024)

Source: IRO Journals, Journal of Ubiquitous Computing and Communication Technologies, Volume 6, Issue 3, September 2024

DIO: IRO Journals – Journal of Ubiquitous Computing and Communication Technologies

Methodology:
The proposed system is an online integrated platform developed to support project management activities through a centralized web-based interface. The platform enables coordination among users by managing project-related information, tracking activities, and facilitating communication. The system architecture focuses on integration, accessibility, and efficient handling of project workflows in a digital environment.

Dataset:
• Dataset Collection: Project-related data including user information, project details, and activity records collected through the online platform.
• Dataset Processing: The data is stored in a centralized database to support integration, monitoring, and management of project activities.

Summary:
The paper "Online Integrated Platform for Project Management" presents a web-based solution aimed at improving project coordination and management through an integrated digital platform. By centralizing project information and workflows, the system enhances accessibility, collaboration, and overall management efficiency. The platform demonstrates how online integration can simplify project handling and support organized execution of project tasks.

Advantages:
• Centralized platform improves coordination and accessibility.
• Online integration supports efficient project monitoring.
• Enhances collaboration among project stakeholders.

Disadvantages:
• Lacks domain-specific focus on academic project evaluation.
• Limited details on evaluation standards and assessment mechanisms.

LITERATURE REVIEW 13

Title: Project Management Information Systems: An Empirical Study of Their Impact on Project Managersand Project Success

Authors: Louis Raymond et al. (2016)

Source: International Journal of Project Management, Volume 26, Issue 2, February 2016

DOI : https://doi.org/10.1016/j.ijproman.2007.06.002

Methodology:
In this empirical study, Raymond et al. examined how Project Management Information Systems (PMIS) affect project managers' tasks and overall project success. The researchers developed a PMIS success model comprising five constructs: PMIS quality, quality of information output, usage level, individual impacts of the PMIS, and project success impacts. They conducted a survey involving 39 project managers across different organizations to collect quantitative data on perceptions of their PMIS tools. Using structured questionnaires, the study measured relationships between system quality, information quality, and performance outcomes, focusing on project planning, scheduling, monitoring, control, and decision-making effectiveness.

Summary:
The study found that high-quality PMIS positively contributes to project management performance by improving managers' ability to plan, organize, monitor, and control project activities. Respondents reported gains in effectiveness and efficiency in managerial tasks, particularly in timelier decision-making, better scheduling, and enhanced monitoring. The PMIS also demonstrated measurable impacts at the organizational level, contributing to improved adherence to budgets, schedules, and technical requirements, which are key metrics for project success. The findings suggest that PMIS use corresponds not only to individual user performance improvements but also to broader project performance benefits.

Advantages:
- Provides empirical evidence connecting PMIS quality to project success outcomes.
- Demonstrates PMIS contribution to planning, scheduling, and control tasks performed by project managers.
- Highlights the importance of information quality for effective decision-making.
- Shows that PMIS use enhances budget control and schedule adherence, key success metrics.
- Offers a theoretical model linking system features to performance, useful for future research.

Disadvantages:
- Based on a small sample size (39 respondents), which may limit generalizability.
- Focuses on managers' perceptions, not objective performance metrics.
- Does not explore longitudinal impacts of PMIS use over time.
- Does not deeply examine specific features or modules within PMIS tools.
- Results are contextual to organizations that already use PMIS, not those transitioning from manual systems.

LITERATURE REVIEW 14

Title: Final Year Supervision Management System as a Tool for Monitoring Computer Science Projects

Authors : Marini Abu Bakar et al. (2017)

Source: Procedia – Social and Behavioral Sciences, Volume 18, 2017

DOI: https://doi.org/10.1016/j.sbspro.2011.05.039

Methodology:

The study by Marini Abu Bakar et al. investigates the challenges in supervising final year projects in the Computer Science programme at Universiti Kebangsaan Malaysia, where students often failed to complete their prototypes on time due to inadequate monitoring mechanisms. The researchers conducted preliminary observations of existing practices, which relied on paper-based log books to document meetings between supervisors and students but lacked any enforcement or automated deadline reminders. Based on this problem identification, a web-based Supervision Management System prototype was designed and developed to support project monitoring and supervision. The prototype comprises modules for user profiles, project monitoring of both software development and report writing stages, and an appointment-setting module to schedule supervisor–student meetings. The study planned to conduct user acceptance testing following system completion.

Summary:

The paper highlights the inadequacy of traditional manual supervision methods used for final year projects, where scheduled deadlines are often not met due to a lack of monitoring triggers in log books. To address this, Marini Abu Bakar et al. propose a web-based supervision management system that centralizes project tracking and improves communication between supervisors and students. By digitizing project monitoring and appointment scheduling, the system aims to provide structure and accountability in the supervision process. While the prototype focuses on three core functional modules, the researchers intended to follow up with user acceptance testing to evaluate its effectiveness in real academic settings.

Advantages:

- Identifies limitations of manual supervision and proposes a digital solution.
- Integrates key supervision tasks like appointment scheduling and project monitoring.
- Provides framework for future extension and evaluation through user testing.
- Contributes early work in automated final year project supervision systems.
- Emphasizes the need for structured, timely supervision support.

Disadvantages :

- Prototype limited to basic functionality; user acceptance testing was planned but not reported in the paper.
- Does not discuss scalability or integration with other academic systems.
- Focuses specifically on one university programme without broader generalization.
- Limited details on technical architecture or performance evaluation.

LITERATURE REVIEW 15

Title: Bots and Insights: Combining Perspectives of Analytics and Software Development in Systems. Analysis and Design Projects

Authors: Madhav Sharma et al. (2025)

Source: Decision Sciences Journal of Innovative Education, 2025

DOI: https://doi.org/10.1111/dsji.70005.

Methodology:

The study adopts a project-based learning approach in which students develop an integrated software system consisting of a chatbot interface and an analytical dashboard connected to a common database. The instructional design follows Merrill's Pebble-in-the-Pond model to ensure structured learning. Multiple user roles interact with the system, and feedback from students is used to evaluate the effectiveness of the approach.
System Features Identified:
Integration of multiple software components into a single platform.
Centralized database shared across modules.
Role-based interaction for different user groups.
Use of intelligent components (chatbot and analytics dashboard)
Real-world software development practices in academic projects.

Summary:

This paper highlights the importance of developing integrated software platforms in academic project environments to enhance student learning and system usability. By combining analytics and intelligent interfaces within a single application, the study demonstrates how complex software systems can be effectively designed and managed by students. The approach emphasizes structured project workflows, multi-user access, and centralized data handling, which contribute to improved project coordination and learning outcomes.
Relevance to Proposed Project:
The concepts presented in this study align with the proposed Academic Project Management System, particularly in terms of centralized project data management, multi-module integration, and role-based system access. Although the paper focuses on instructional design rather than full-scale project monitoring, it supports the need for integrated digital platforms to manage and evaluate academic software projects efficiently.

Advantages:

- Encourages development of integrated, real-world software systems
- Demonstrates effective use of centralized databases in academic projects
- Supports role-based system design for multiple stakeholders
- Enhances coordination and monitoring of academic software development

Disadvantages:

- Does not address full project lifecycle management or approval workflows
- Lacks features related to plagiarism detection and rubric-based evaluation
- Focuses more on learning outcomes than institutional project tracking